WORKED SOLUTION

STEP III 1996, Question 10

Mechanics • Equilibrium with Friction

STEP III 1996, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q10

The key idea

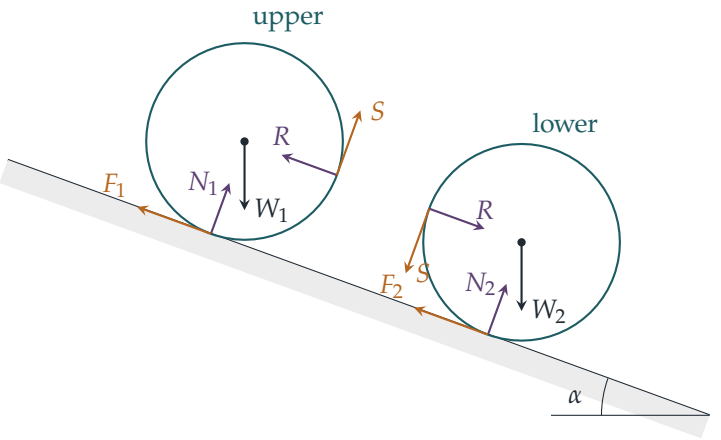
Write down the six equilibrium equations (two resolutions and one moment equation for each cylinder) and you will find that they determine every contact force uniquely. The moment equations are what make it work: each cylinder has just two frictional forces with a moment about its centre, so these must be equal in size. After that, each of the four inequalities is simply "friction $\leq$ coefficient $\times$ normal reaction" at one of the three lines of contact, plus the condition that the push between the cylinders is not negative.

Solution

Model. Each cylinder is a uniform rigid body of radius a, so its weight acts at its centre (on the axis). Work in a vertical cross-section through the centres. Since the radii are equal and both cylinders rest on the plane, the line of centres is parallel to the plane, and the contact between the cylinders is at the midpoint of that line. Take the x-direction *down* the slope and the y-direction perpendicular to the plane, pointing out of it.

Define the forces (all components may turn out to be of either sign):

- N_1, N_2: normal reactions of the plane on the upper and lower cylinders;
- F_1, F_2: frictional forces of the plane on them, measured *up* the slope;
- R: the normal force between the cylinders, along the line of centres, pushing the lower cylinder down the slope and the upper one up the slope;
- S: the frictional force between the cylinders, perpendicular to the plane, acting *away from* the plane on the upper cylinder and so *towards* the plane on the lower one (by Newton's third law).



(For clarity the cylinders are drawn apart; R and S act at the single line of contact. Each weight acts vertically downwards through the centre.)

Moments. Take moments about the centre of each cylinder. The normal forces N_1, N_2, R and the weights all pass through the centres, so only the frictional forces have moments. Each frictional force acts at distance a from the centre, tangentially. For the upper cylinder, F_1 (up the slope, at the bottom of the cylinder) and S (away from the plane, at its downhill side) have opposite moments, giving

$$aF_1 = aS, \qquad \text{so} \qquad F_1 = S.$$

In the same way, for the lower cylinder F_2 (up the slope, at its bottom) and S (towards the plane, at its uphill side) have opposite moments, so $F_2 = S$. Hence

$$F_1 = F_2 = S.$$

Ask yourself Why must the friction from the plane on each cylinder equal the friction between the cylinders?

Because a cylinder that is free to roll is only held in equilibrium if the couples about its axis cancel, and friction forces are the only forces with a moment about the axis. You can check the senses: on the upper cylinder, the plane's friction up the slope at the bottom would roll it downhill, i.e. clockwise in the diagram, while an upward frictional force on its downhill side turns it anticlockwise.

Resolving. Parallel to the plane (positive down the slope):

$$\begin{aligned} \text{upper:} \quad & W_1 \sin\alpha - F_1 - R = 0, \\ \text{lower:} \quad & W_2 \sin\alpha - F_2 + R = 0. \end{aligned}$$

Adding, and using $F_1 = F_2 = S$,

$$F_1 = F_2 = S = \tfrac{1}{2}(W_1 + W_2)\sin\alpha,$$

which is half the total down-slope component of weight, as you would expect. Subtracting then gives

$$R = \tfrac{1}{2}(W_1 - W_2)\sin\alpha.$$

Perpendicular to the plane (positive outwards):

$$\begin{aligned} \text{upper:}\quad & N_1 - W_1\cos\alpha + S = 0 \quad \Longrightarrow \quad N_1 = W_1\cos\alpha - \tfrac{1}{2}(W_1 + W_2)\sin\alpha, \\ \text{lower:}\quad & N_2 - W_2\cos\alpha - S = 0 \quad \Longrightarrow \quad N_2 = W_2\cos\alpha + \tfrac{1}{2}(W_1 + W_2)\sin\alpha. \end{aligned}$$

So the six equations fix all six forces. Equilibrium is possible exactly when these forces are physically allowed: every normal reaction is non-negative and every frictional force is at most the coefficient of friction times the corresponding normal reaction.

Part (i)

The cylinders can only push on each other, so $R \geq 0$. Since $\sin\alpha > 0$, this gives $W_1 \geq W_2$.

Part (ii)

At the contact between the cylinders we need $S \leq \mu R$:

$$\tfrac{1}{2}(W_1 + W_2)\sin\alpha \leq \mu \cdot \tfrac{1}{2}(W_1 - W_2)\sin\alpha.$$

The left-hand side is strictly positive, so this forces $R > 0$, i.e. $W_1 > W_2$ (equal weights are in fact impossible). Dividing by the positive quantity $\tfrac{1}{2}(W_1 - W_2)\sin\alpha$,

$$\mu \geq \frac{W_1 + W_2}{W_1 - W_2},$$

as required.

Part (iii)

At the upper cylinder's contact with the plane we need $N_1 \geq 0$ and $F_1 \leq \mu_1 N_1$. Since $F_1 > 0$, the second condition already forces $N_1 > 0$. Dividing $F_1 \leq \mu_1 N_1$ by $\tfrac{1}{2}(W_1 + W_2)\sin\alpha > 0$:

$$1 \leq \mu_1\left(\frac{2W_1\cos\alpha}{(W_1 + W_2)\sin\alpha} - 1\right) = \mu_1\left(\frac{2W_1\cot\alpha}{W_1 + W_2} - 1\right).$$

The bracket is $N_1 / \left(\tfrac{1}{2}(W_1 + W_2)\sin\alpha\right)$, which is positive, so we may divide by it:

$$\mu_1 \geq \left(\frac{2W_1\cot\alpha}{W_1 + W_2} - 1\right)^{-1}.$$

(Equilibrium also needs the bracket to be positive, i.e. $\tan\alpha < 2W_1/(W_1 + W_2)$; otherwise the upper cylinder would be lifted off the plane.)

The inequality for μ_2

Here $N_2 = W_2\cos\alpha + \frac{1}{2}(W_1+W_2)\sin\alpha$ is automatically positive, and $F_2 \le \mu_2 N_2$ becomes, after dividing by $\frac{1}{2}(W_1+W_2)\sin\alpha$,

$$1 \le \mu_2\left(\frac{2W_2\cot\alpha}{W_1+W_2}+1\right), \qquad \text{i.e.} \qquad \mu_2 \ge \left(\frac{2W_2\cot\alpha}{W_1+W_2}+1\right)^{-1}.$$

Ask yourself Why is the sign $+1$ for the lower cylinder but -1 for the upper one?

The friction S between the cylinders pulls the upper cylinder away from the plane and pushes the lower one into it. So the plane presses less on the upper cylinder and more on the lower one, and the upper cylinder needs a larger ratio of friction to normal reaction.

Checks

- Adding the two cylinders' equations gives the whole system: total plane friction $(W_1+W_2)\sin\alpha$ up the slope and total normal reaction $(W_1+W_2)\cos\alpha$, exactly as for one rigid block.
- As $\alpha \to 0$ all frictional forces and R tend to 0, and both bounds on μ_1, μ_2 tend to 0, as they should on level ground.
- Numerically, with $W_1 = 3$, $W_2 = 1$, $\alpha = 20°$: solving the six equations directly gives $R \approx 0.342$, $S = F_1 = F_2 \approx 0.684$, $N_1 \approx 2.135$, $N_2 \approx 1.624$, so $\mu \ge 2$, $\mu_1 \ge 0.320$ and $\mu_2 \ge 0.421$, agreeing with the formulae. The same agreement was checked for many random choices of W_1, W_2 and α.

Common slips

- Forgetting the moment equations. Without them there are six unknowns and only four equations, and the problem looks indeterminate.
- Getting the direction of the friction between the cylinders wrong, which swaps the ± 1 in the last two answers.
- Overlooking that the lighter cylinder must be the lower one: if the upper cylinder were lighter, R would have to be negative.

Beyond STEP

This is a statically determinate system: the equilibrium equations fix every force, so the friction laws become checkable conditions. When there are more contacts than equations (a ladder against a rough wall on rough ground, say) the system is statically

indeterminate, and rigid-body statics alone cannot say how the forces are shared. Engineers then bring in the elasticity of the bodies. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers rigid-body equilibrium and torque carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.